

Tafila Technical University

Faculty of Information Technology



Course Name: Digital Forensics.

Project Title: Lone Wolf.

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1. Introduction:

In light of the rapid and massive digital advancements of our time, crimes no longer occur solely in the physical world but have expanded into the digital realm. This makes digital forensics a crucial tool for uncovering and investigating these crimes and suspicious activities. Digital forensics relies on proficiency in using investigative tools, analyzing digital evidence, and the investigator's ability to connect digital evidence to reveal suspicious activities.

This project examines a digital forensics analysis of a case known as the "lone wolf" scenario. The objective of the investigation is to analyze a digital forensic image taken from a suspect's device.

The scenario, "Lone Wolf 2018," is based on material obtained from a simulated seizure of a laptop belonging to a fictitious individual who was allegedly planning a mass shooting. In this scenario, the fictitious individual's brother reported his increasingly disturbing behavior to the police. As a result of this report, the police seized the brother's laptop and then copied its data using FTK Imager software.

When analyzing digital evidence, the focus was on:

- 1- Recovering deleted images: to access files, documents, and photos that the suspect attempted to hide.
- 2- Web history and search analysis: to identify the websites visited and searched for by the suspect and to understand their interests.
- 3- Keyword search and user activity analysis: to extract hidden text and analyze user activity.

2. Lab Setup:

2.1 virtual environment:

For this project, a virtual machine environment with a guest operating system of Windows 10 was used.

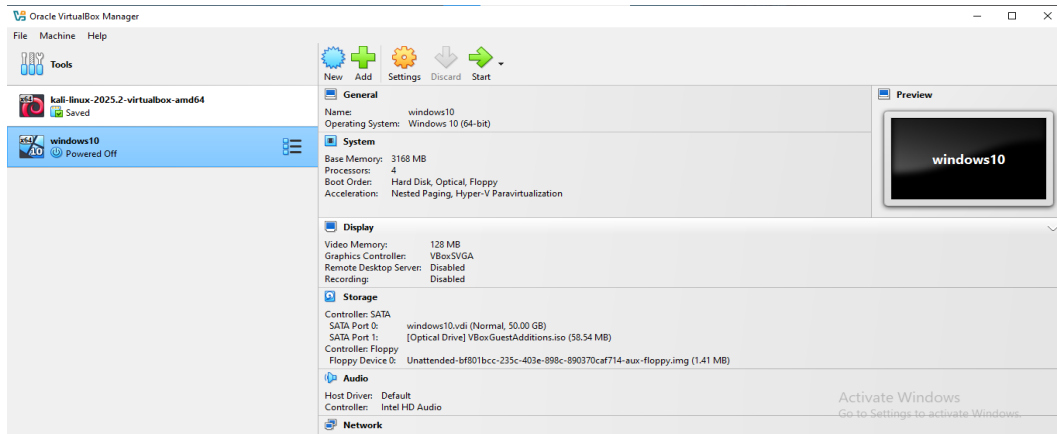


Figure 1: Virtual machine settings information

2.2 Tool Installation:

1-FTK Imegar: Used for verifying the integrity of the E01 image files and checking the MD5/SHA1 hashes.

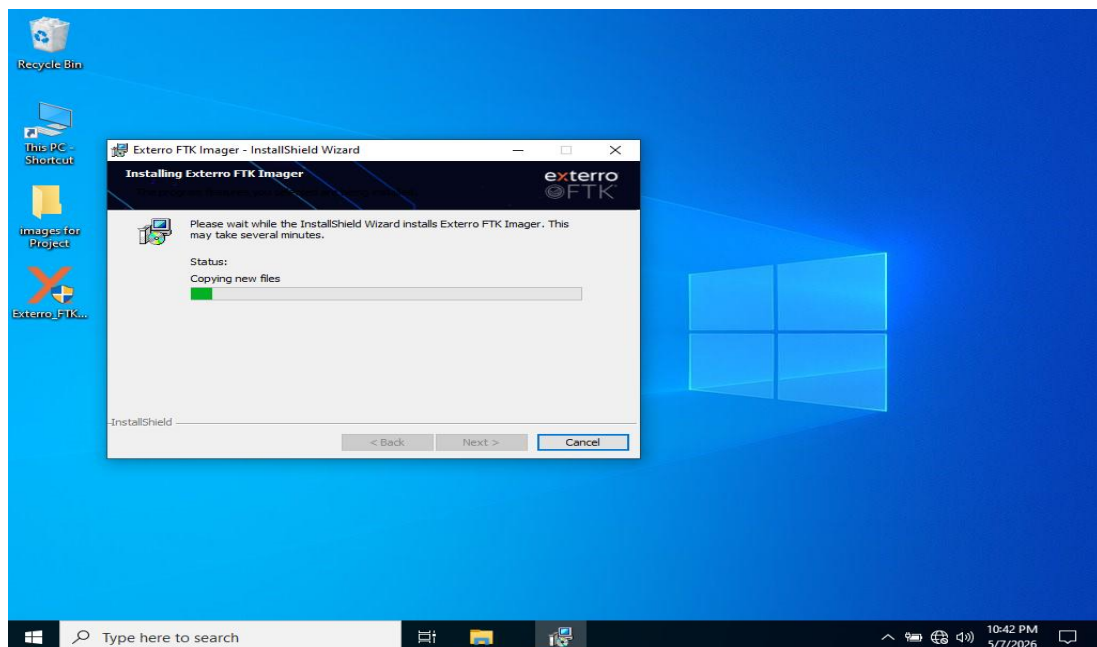


Figure 2: Install FTK Imager software

2-Autopsy: the premier end-to-end open source digital forensics platform. used for disk analysis, data carving, and web history recovery.

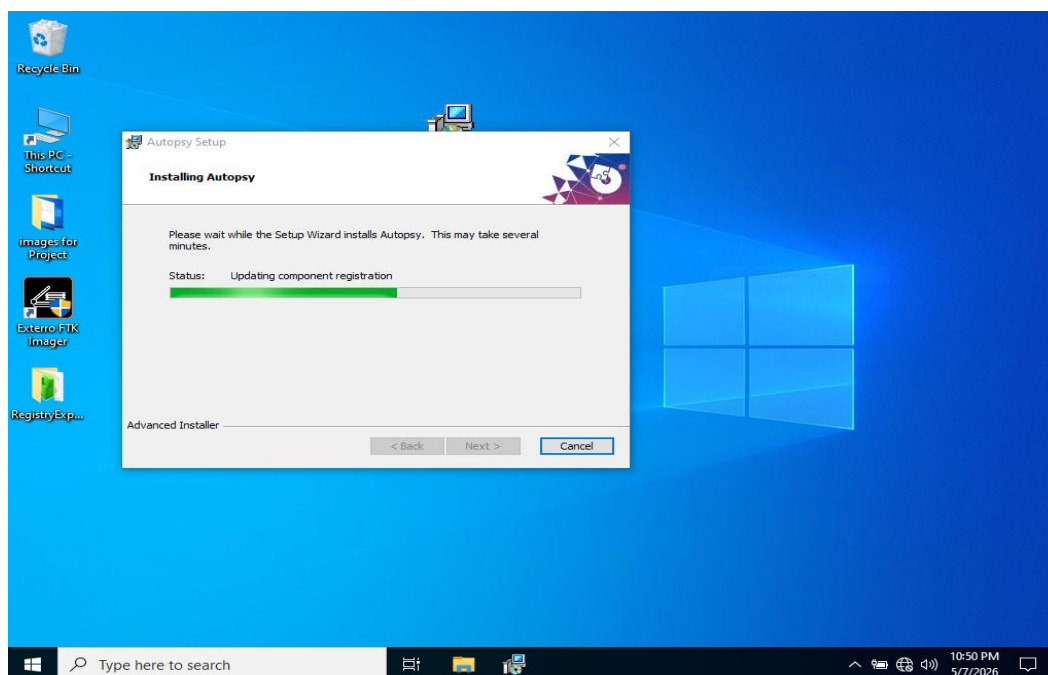


Figure 3: Installing Autopsy software

3. Evidence Acquisition:

3.1 Evidence Description:

The evidence provided for this project consists of a forensic digital image of a laptop computer belonging to a fictitious person who was planning a mass shooting. The image is stored in E01 format. The evidence is divided into 9 fragmented files (LoneWolf.E01 to LoneWolf.E09).

3.2 Hash Verification:

To ensure the integrity of digital evidence and prevent tampering, a hashing verification process was performed using FTK Imager software.



Figure 4: Hash Verification using FTK Imager

4. Forensic Analysis:

Task 1: Deleted Files & Images Recovery:

The focus was on recovering deleted media using the PhotoRec Carver module, and through analyzing the recovered media, important evidence was discovered, including pictures and files of firearms and maps, indicating that the suspect had deliberately deleted them. A deleted image was also discovered in the Recycle Bin belonging to the National Democratic Party headquarters. This evidence indicates prior planning and reconnaissance by the suspect for a possible attack.

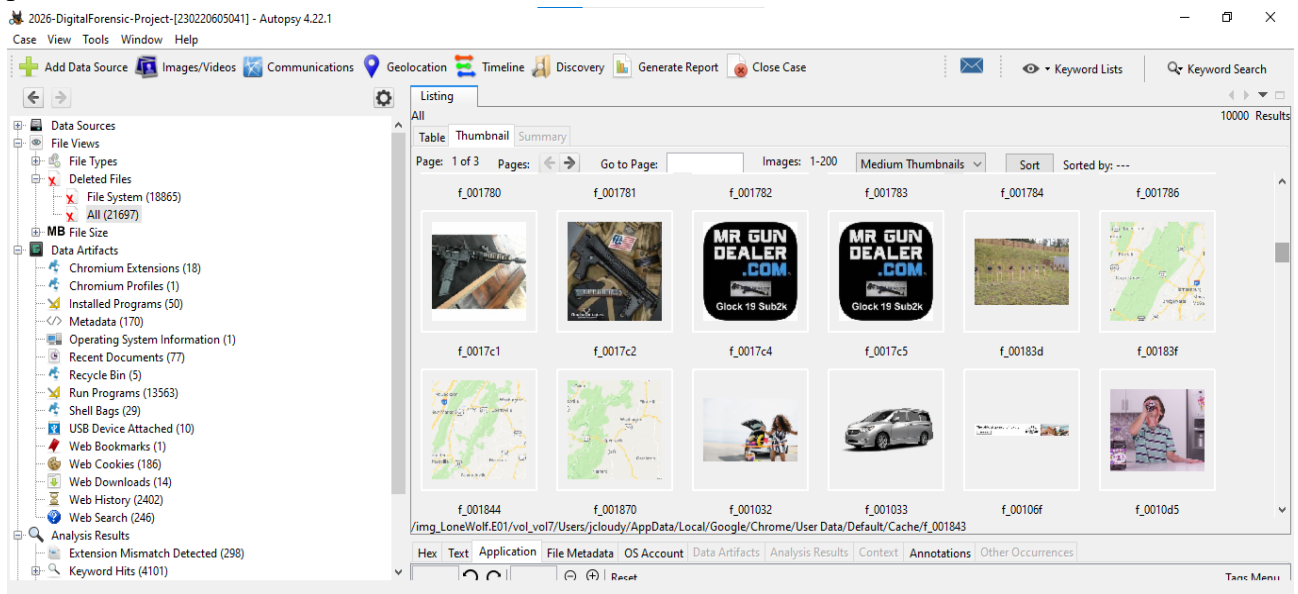


Figure 5: Overview of Recovered Media

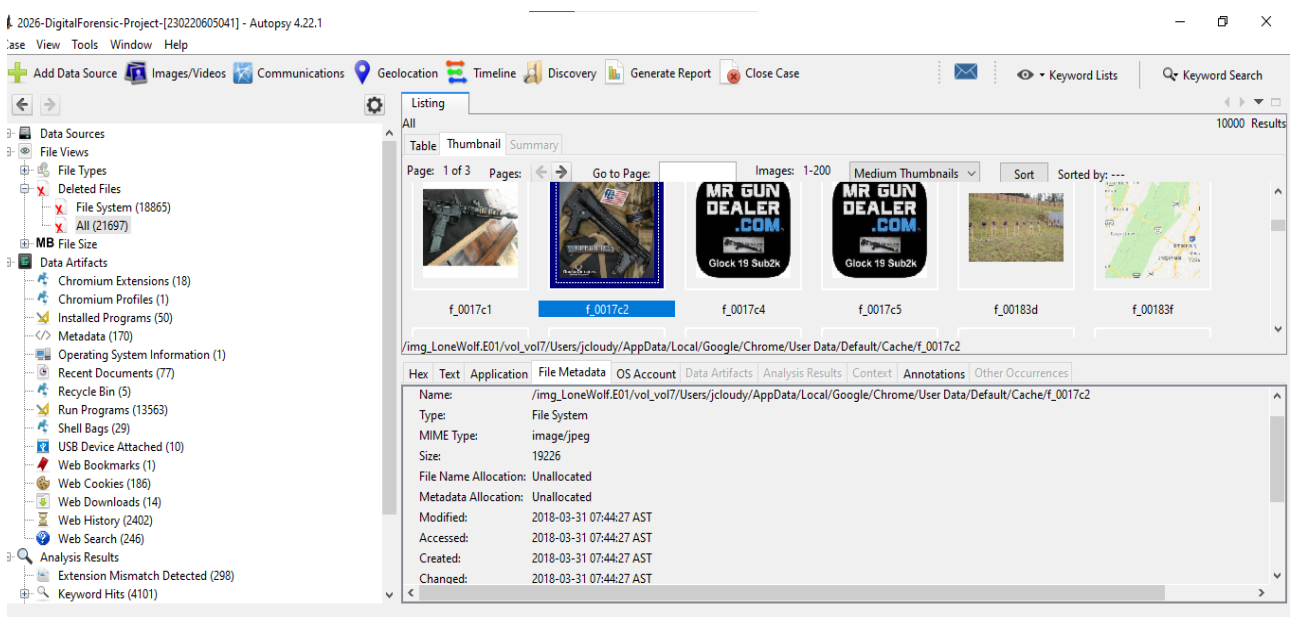


Figure 6: Weapon image metadata

Extracted image:



Figure 7: Image extracted of the National Democratic Party headquarters

Task 2: Web History Analysis:

In the second task, the investigation focused on the suspect's online activity to link the recovered images and files to his browsing history and what he was searching for on the browser. Important evidence was discovered, as it was found that the suspect used the browser to search for and inquire about the website of the National Democratic Headquarters. He was also searching for nearby gun shops, weapons and ammunition, and a shooting range, indicating that he wanted to practice with the weapons he had obtained.

Web History							2402 Results
Table Thumbnail Summary							This is a DataResult window
							Save Table as CSV
Source Name	S	C	O	URL	Date Accessed	Referrer URL	
History			0	https://www.google.com/search?rlz=1C1CHBF_enUS7...	2018-03-30 02:17:27 AST	https://www.google.com/search?rlz=1C1CHBF_enU	
History			0	https://www.dailykos.com/story/2017/4/17/1653154/-...	2018-03-30 02:17:56 AST	https://www.dailykos.com/story/2017/4/17/1653154	
History			0	http://www.dailykos.com/stories/2017/4/17/1653154/...	2018-03-30 02:17:56 AST	http://www.dailykos.com/stories/2017/4/17/1653154	
History			0	https://www.dailykos.com/stories/2017/4/17/1653154...	2018-03-30 02:17:56 AST	https://www.dailykos.com/stories/2017/4/17/165315	
History			0	https://www.google.com/search?rlz=1C1CHBF_enUS7...	2018-03-30 02:18:40 AST	https://www.google.com/search?rlz=1C1CHBF_enU	
Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences							
Result: 311 of 2534 Result							Web History
Visit Details							
Title: democratic national headquarters - Google Search							
Username: Default							
Date Accessed: 2018-03-30 02:18:40 AST							
Domain: google.com							
URL: https://www.google.com/search?rlz=1C1CHBF_enUS790US790&biw=1366&bih=613&tbnm=isch&sa=1&ei=j3O9Ws2xEcyWjwPqLLrwDQ&q=democratic+							
Referrer URL: https://www.google.com/search?rlz=1C1CHBF_enUS790US790&biw=1366&bih=613&tbnm=isch&sa=1&ei=j3O9Ws2xEcyWjwPqLLrwDQ&q=democratic+							
Program Name: Google Chrome							

Figure 8: web history for searching about Democratic National Headquarters

Web History							2402 Results
Table Thumbnail Summary							Save Table as CSV
Source Name	S	C	O	URL	Date Accessed	Referrer URL	
History			0	https://www.gunbroker.com/Pistol-Ammunition/sear...	2018-03-31 07:38:12 AST	https://www.gunbroker.com/Pistol-Ammunition/se	
History			0	https://www.gunbroker.com/Pistol-Ammunition/sear...	2018-03-31 07:38:12 AST	https://www.gunbroker.com/Pistol-Ammunition/se	
History			0	https://www.gunbroker.com/Pistol-Ammunition/sear...	2018-03-31 07:38:13 AST	https://www.gunbroker.com/Pistol-Ammunition/se	
History			0	https://www.gunbroker.com/Pistol-Ammunition/sear...	2018-03-31 07:38:13 AST	https://www.gunbroker.com/Pistol-Ammunition/se	
History	▼		0	https://www.gunbroker.com/search?Keywords=9mm...	2018-03-31 07:38:19 AST	https://www.gunbroker.com/search?Keywords=9m	▼

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Result: 594 of 2534 Result <=>									

Visit Details	
Title:	9mm rifle For Sale - Buy 9mm rifle Online at GunBroker.com
Username:	Default
Date Accessed:	2018-03-31 07:38:19 AST
Domain:	gunbroker.com
URL:	https://www.gunbroker.com/search?Keywords=9mm+rifle&BuyNowOnly=1&Sort=5&PageSize=24
Referrer URL:	https://www.gunbroker.com/search?Keywords=9mm+rifle&BuyNowOnly=1&Sort=5&PageSize=24
Program Name:	Google Chrome

Figure 9: web history for searching about 9mm rifle for sale

Web History							2402 Results
Table Thumbnail Summary							Save Table as CSV
Source Name	S	C	O	URL	Date Accessed	Referrer URL	
History	▼		0	https://www.google.com/search?rlz=1C1CHBF_enUS7...	2018-03-30 02:18:40 AST	https://www.google.com/search?rlz=1C1CHBF_enU	
History			0	https://www.google.com/search?rlz=1C1CHBF_enUS7...	2018-03-30 02:18:54 AST	https://www.google.com/search?rlz=1C1CHBF_enU	
History			0	https://www.google.com/search?rlz=1C1CHBF_enUS7...	2018-03-30 02:18:59 AST	https://www.google.com/search?rlz=1C1CHBF_enU	
History	▼		0	https://www.google.com/search?q=430+south+capit...	2018-03-30 02:19:03 AST	https://www.google.com/search?q=430+south+cap	
History			0	https://www.google.com/maps/place/430+S+Capitol...	2018-03-30 02:19:08 AST	https://www.google.com/maps/place/430+S+Capit	▼

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Result: 314 of 2534 Result <=>									

Visit Details	
Title:	430 south capital street, DC - Google Search
Username:	Default
Date Accessed:	2018-03-30 02:19:03 AST
Domain:	google.com
URL:	https://www.google.com/search?q=430+south+capitol+street,+DC&rlz=1C1CHBF_enUS790US790&source=Inms&sa=X&ved=0ahUKEwjT15b1ZLaAhVh
Referrer URL:	https://www.google.com/search?q=430+south+capitol+street,+DC&rlz=1C1CHBF_enUS790US790&source=Inms&sa=X&ved=0ahUKEwjT15b1ZLaAhVh
Program Name:	Google Chrome

Figure 10: web history for searching about Democratic National Headquarters

Web Search

246 Results

TableThumbnailSummary

Save Table as CSV

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History				google.com	democratic offices in DC	Google Chrome	2018-03-30 02:17:27 AST	LoneWolf.E01
History	▼			google.com	democratic national headquarters	Google Chrome	2018-03-30 02:18:40 AST	LoneWolf.E01
History	▼			google.com	430 south capital street	Google Chrome	2018-03-30 02:18:54 AST	LoneWolf.E01
History	▼			google.com	430 south capital street, DC	Google Chrome	2018-03-30 02:18:59 AST	LoneWolf.E01
History				google.com	430 south capitol street, DC	Google Chrome	2018-03-30 02:19:03 AST	LoneWolf.E01

HexTextApplicationSource File MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Result: 2338 of 2534Result

Web Search

Term: 430 south capital street

Time: 2018-03-30 02:18:54 AST

Domain: google.com

Program Name: Google Chrome

Figure 11: web search about Democratic National Headquarters

Listing						
Web History						
Table Thumbnail Summary						
Save Table as CSV						
Source Name	S	C	O	URL	Date Accessed	Referrer URL
History	▼		0	https://www.google.com/search?q=ls+there+a+map+...	2018-03-30 02:13:25 AST	https://www.google.com/search?q=ls+there+a+ma
History	▼		0	https://maps.google.com/maps?q=ls+there+a+map+...	2018-03-30 02:13:20 AST	https://maps.google.com/maps?q=ls+there+a+ma
History	▼		0	https://www.google.com/maps?q=ls+there+a+map+...	2018-03-30 02:13:21 AST	https://www.google.com/maps?q=ls+there+a+map
History			0	https://www.google.com/maps?q=ls+there+a+map+...	2018-03-30 02:13:21 AST	https://www.google.com/maps?q=ls+there+a+map
History			0	https://www.google.com/maps/place/National+Rifle...	2018-03-30 02:13:24 AST	https://www.google.com/maps/place/National+Rifl
Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences						
Result: 296 of 2534 Result						
Web History						
Visit Details						
Title: Is there a map of gun free zones - Google Search						
Username: Default						
Date Accessed: 2018-03-30 02:13:25 AST						
Domain: google.com						
URL: https://www.google.com/search?q=ls+there+a+map+of+gun+free+zones&rlz=1C1CHBF_enUS790US790&oq=ls+there+a+map+of+gun+free+zones&						
Referrer URL: https://www.google.com/search?q=ls+there+a+map+of+gun+free+zones&rlz=1C1CHBF_enUS790US790&oq=ls+there+a+map+of+gun+free+zones&						
Program Name: Google Chrome						

Figure 12: web history for searching about a map of gun free zones

Task 3: Keyword search and user activity analysis:

The final stage of the analysis involved keyword research and an examination of system data and files, where the most important clue was discovered. The login name was 'jcloudy', and a Microsoft Word file named Planning.docx was found containing the suspect's detailed plan, divided into:

1. Target:

- 1.1 Must have good escape route
- 1.2 Preferably near Airport
- 1.3 Must be Gun Free zone.

2. Supplies:

2.1 Gun (black market):

- 2.1.1 Norther VA Gun Works 7518 Fullerton Rd # K, Springfield, VA 22153
- 2.1.2 NOVA 412 W Broad Street Falls Church, VA 22046

2.2 Ammo:

- 2.2.1 9mm is 1000 for \$360
- 2.2.2 Kel-Tec Sub 2000 9mm \$400.

2.3 Latex gloves

2.4 Velcro tear away clothing?

2.5 Cash

3. Escape:

3.1 No Extradition countries

3.1.1 Indonesia (Nicer, but more expensive)

3.1.2 Vietnam

3.1.3 Can live very well on 100 a day, for 9 years.

3.2 Buy tickets for same day

3.3 Preferable direct flight

3.4 Have suitcase in car.

4. Release:

4.1 Start writing ideas and thoughts

4.2 Save to separate locations for redundancy

4.3 Place it in the cloud for remote access

4.4 “Press Release” once home free.

This plan proves that the suspect's activities were not merely random searches, but rather part of a plan and conspiracy to commit a crime and then flee the country.

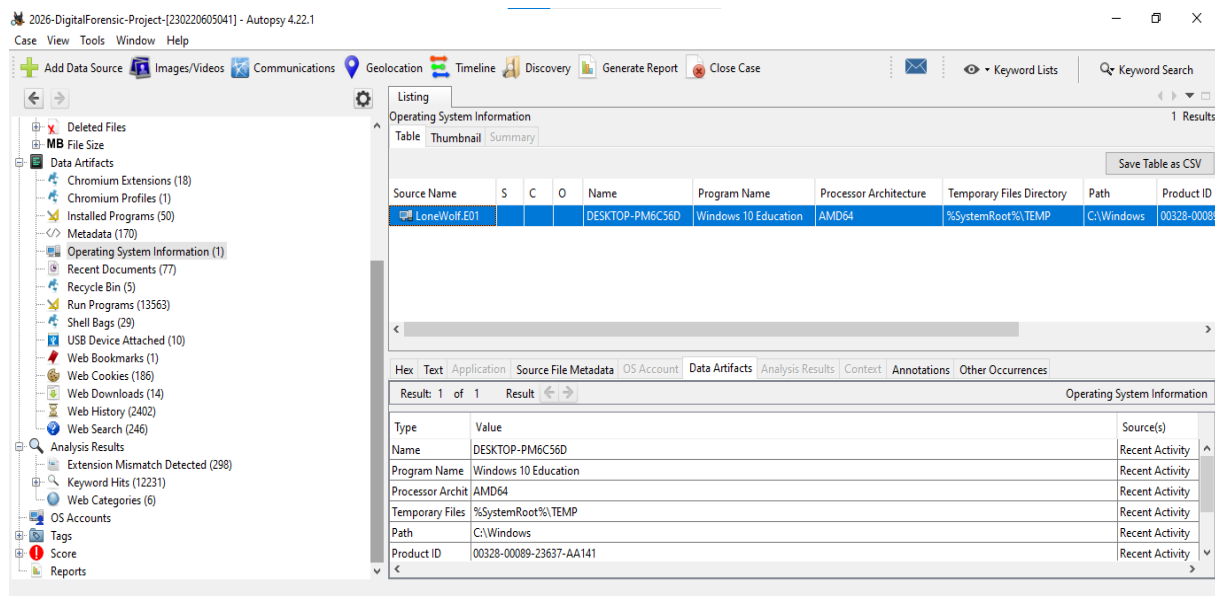


Figure 13: OS Accounts information

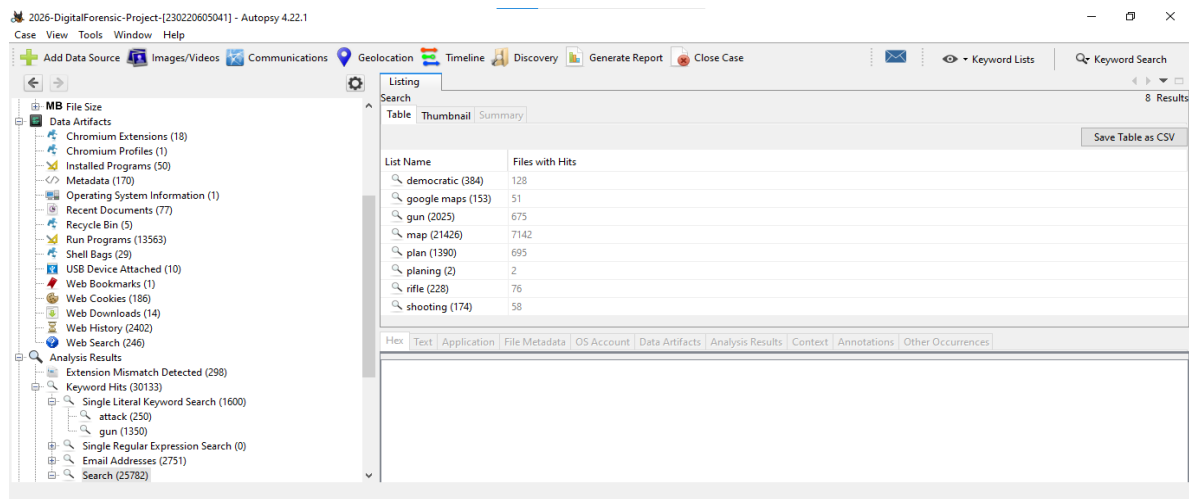


Figure 14: Keyword Search Hits

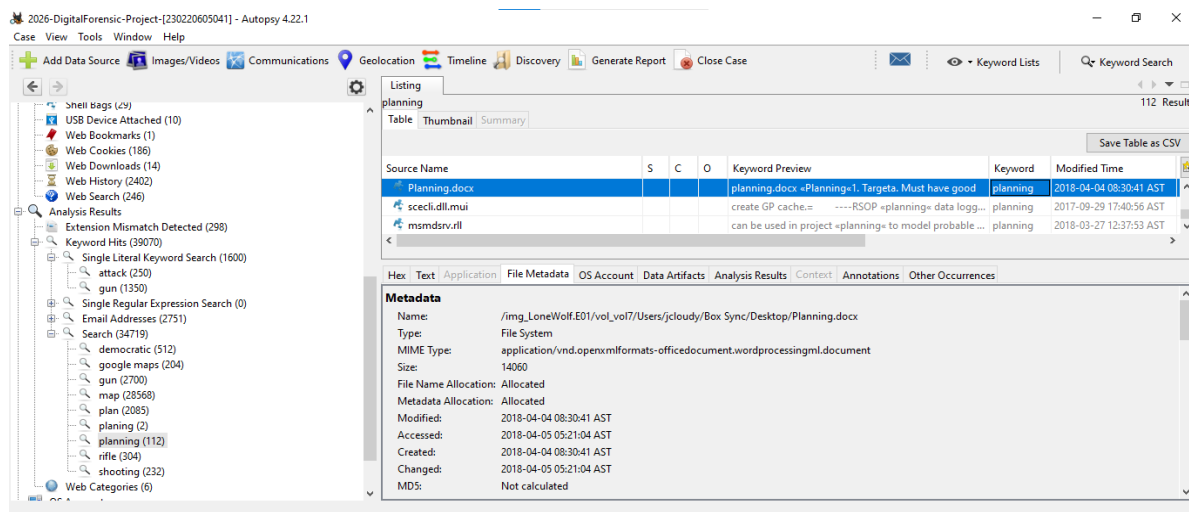


Figure 15: Planning file metadata

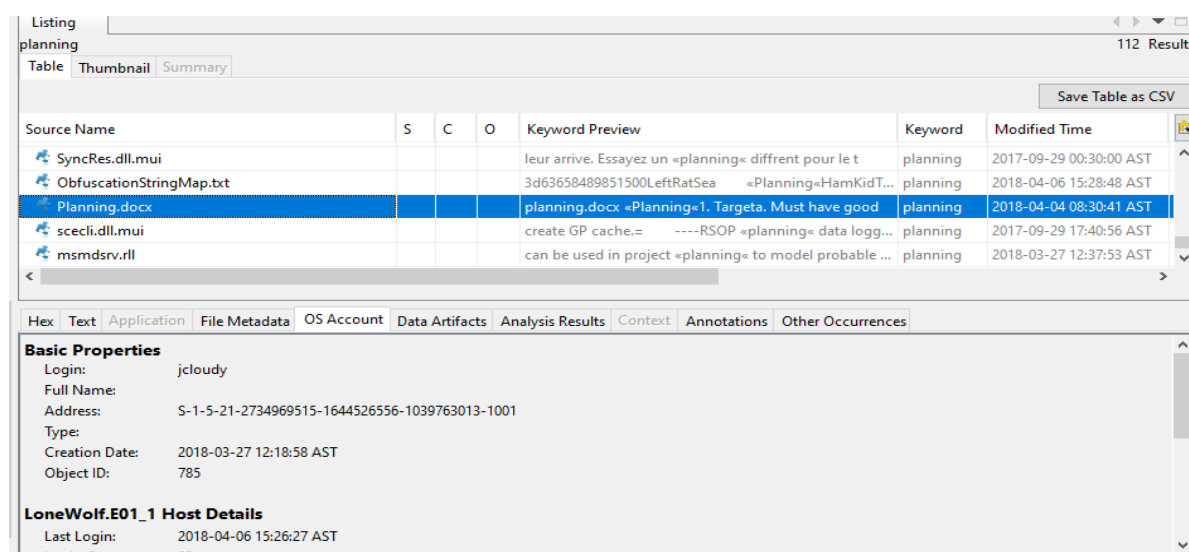


Figure 16: User account

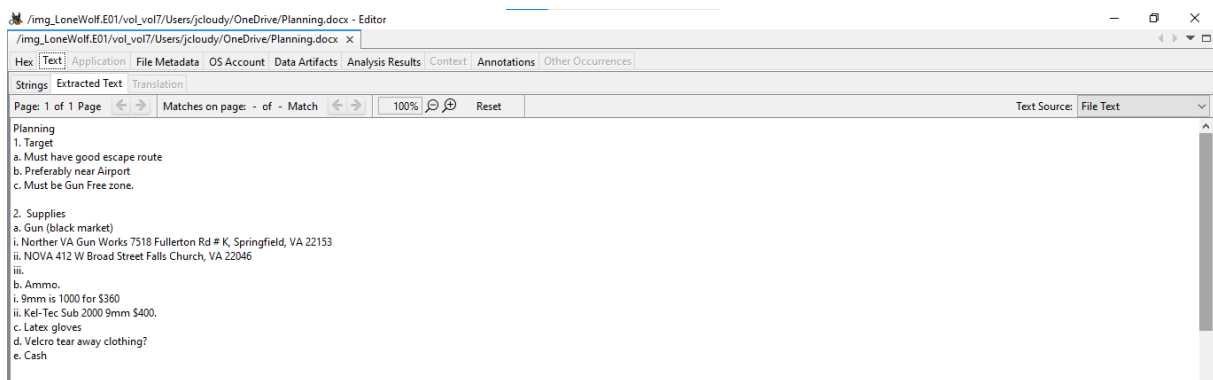


Figure 17: Planning file content part 1

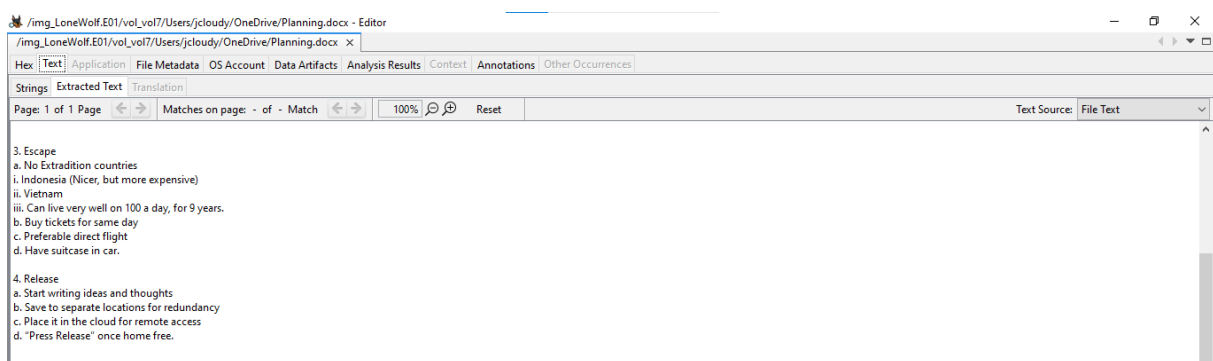


Figure 18: Planning file content part 2

Timeline Analysis (Reconstructing the Events):

A chronological reconstruction analysis was conducted to clarify the suspect's behavior, revealing a significant increase in activity between March 26 and April 6, 2018. This included the suspect searching for a nearby shooting range.

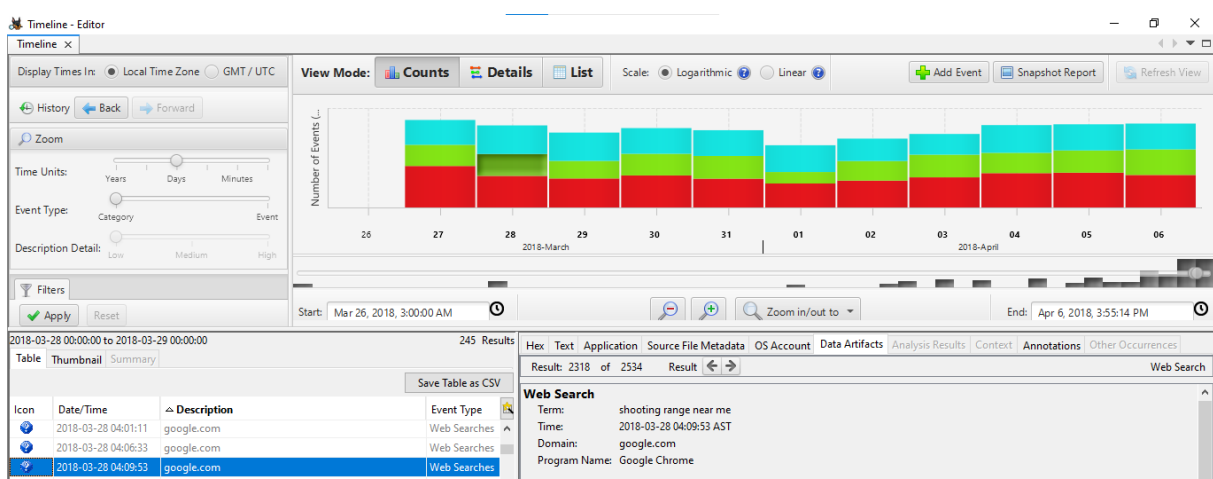


Figure 19: Timeline

5. Mitigation Strategies:

To prevent similar "Lone Wolf" threats and enhance digital security posture, the following mitigation strategies are recommended based on the findings of this investigation:

1. Proactive Monitoring and Early Warning Systems:

Keyword Triggering: Implementing advanced monitoring systems within corporate or public networks that flag specific "high-risk" keywords (e.g., "mass shooting," "black market weapons," "police response times") to security administrators.

Endpoint Detection and Response (EDR): Deploying EDR solutions on organization-owned devices to detect unusual patterns of behavior, such as accessing sensitive geographical data or creating attack blueprints.

2. Cloud Security:

Storage Monitoring: Since the suspect used OneDrive for storing the Planning.docx file, organizations should implement DLP policies that scan cloud storage for suspicious documents or sensitive keywords before they can be synchronized or shared.

Shadow IT Control: Restricting the use of unauthorized cloud services that could be used to hide criminal planning from local investigators.

3. Network and Physical Security Correlation:

Physical Security Awareness: Increasing security presence at "Gun-Free Zones" and political headquarters that may be targeted, as identified in the suspect's research history.

Geofencing Alerts: Implementing alerts for attempts to access detailed blueprints or floor plans of sensitive public buildings without proper authorization.

4. Forensic Preparedness and Incident Response:

Regular Backups and Logging: Ensuring that system logs (Event Logs, Registry, Browser History) are preserved and backed up. This allows forensic investigators to reconstruct timelines more effectively as seen in this case.

5. User Awareness Training:

Educating individuals (like the suspect's brother in the scenario) on how to recognize and report suspicious digital or behavioral patterns to authorities before an incident occurs.

6. Results & Conclusion:

After conducting a comprehensive digital forensic analysis of the digital forensic images "Lone Wolf" and after performing the process of recovering deleted images and files, analyzing Web History and Web search, and performing Keyword search and user activity analysis, evidence was found that included images of weapons, images of the National Democratic Party headquarters, and evidence of the suspect's browser search records for shooting ranges, gun shops, weapons and ammunition, the location of the National Democratic Party headquarters, and the police response. Through Keyword search and system file analysis, a file related to the suspect's plan named Planning.docx was found, containing details of the suspect's plan. The existence of this evidence confirms that this was an operation that the suspect prepared with the intention of committing a criminal act.

Appendix : Chain of custody documentation:

Case Number:	2026-Case-01
Data of acquisition:	2026-05-01
Date of Investigation:	2026-05-07
Description:	Forensics Image "Lone Wolf"
MD5 Hash value:	7af48fa65519e84246b1729e5b68f140
From:	Digital Corpora
Examiner:	Abdelrahman Mohammad Hdaib
PURPOSE OF TRANSFER:	Transfer for forensic analysis